



Animal Welfare
Farmed Animal Welfare
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Forecasts estimate limited cultured meat production through 2050

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Note: For brevity, we use M for a million (10^6) and B for a billion (10^9).

Summary

- We (Neil and Linch) developed forecasting questions around cultured meat reaching annual production volume sold in metric tons ($>100,000$, $>1M$, $>10M$, $>50M$) by a certain year (2031, 2036, 2051) in addition to hypothesized signposts of progress (funding, researchers, input costs, food service sales, and public support). For context, the annual production

and the annual production of [seaweed](#) in 2010 was 200,000 metric tons.

- Here we present an initial set of forecasts from a panel of paid forecasters. We plan to expand forecasting on similar questions in a [Metaculus tournament](#) so that we can see how forecasts are affected by news of supposedly important breakthroughs.
- Despite some variation, **the majority of probabilities were for low production volumes.** The aggregated probabilities from our panel include a 54% probability that less than 100,000 metric tons of cultured meat (where >51% of the “meat” is produced directly from animal cells) will be produced and sold at any price in a 12-month period before the end of 2051.

Aggregated probabilities of cultured meat production targets

Metric tons	2031	2036	2051
>100,000	15%	22%	46%
>1M	3%	9%	31%
>10M	Not asked	3%	18%
>50M	Not asked	Not asked	9%

- These results suggest that rather than pursuing strategies that assume a small nudge can make cultured meat widely available in the near term, there may be more benefit in ensuring long-term support for the industry or in alternatives to cultured meat.
- Engineering new types of bioreactors, building out supply chains of key ingredients, securing broad political support for public funding, and establishing a pipeline of researchers seem promising ways to nudge production trajectories upwards over a timeframe of a century, not decades.
- This report is about describing uncertainty and models, not justifying any specific probability. The main cruxes of disagreement appear to stem from

systems, sources of reference classes, and how much to anchor based on the estimates in the Humbird ([2020](#)) techno-economic analysis.

Introduction

Is it worth the effective altruism (EA) community trying to accelerate the growth of cultured meat production? Should EA just let market forces move it forward? Should EA invest directly in cultured meat R&D or identify high-leverage ways to increase funding? Or should EA just not invest in it because it is insufficiently promising?

A key question is whether (and how much) EA philanthropy and advocacy can counterfactually increase the probability that cultured meat becomes a significant part of the food supply. Imagine we expect an X% probability that at least 50M metric tons of meat and seafood production in 2051 will come from cultured meat (an arbitrary but significant threshold we estimate to be 5% to 7% of total meat demand by then). Can we see some ways to increase that probability to a multiple of X through targeted grants? That may be something that the market would not be incentivized to do (e.g., because it is too high risk, or venture capitalists may think they are unable to capture most of the value), but that EA funders could fill (due to our longer time horizons, risk neutrality, and moral preferences).

- If cultured meat production can scale to a significant portion of meat production in a reasonable timeframe (arbitrarily >50M metric tons in 30 years), it might be worth spending more on it now: pouring money into open access research prizes, pushing for partisan government spending wherever it is most tractable today, creating positive news coverage, convincing biotech PhDs to focus on cultured meat topics, and encouraging animal advocacy groups to include it in their plans.
- If >50M metric tons of cultured meat is only possible on longer timeframes but still better than the alternatives, then maybe we should invest in things with long-term payoffs: building out the academic field of cultured meat more, building broad political support support for funding in countries where most meat demand will be, funding biotech research projects that

remaining in factory farms in the meantime.

- If it seems like >50M metric tons is not viable at all, and cultured meat is limited to a minor role, then we might want to not invest resources in it at all relative to other options.

Unfortunately, cultured meat seems to be one of those technologies where hype clouds accurate assessments of the state of the industry. Cultured meat companies' track record of making public predictions appears to be systematically overoptimistic ([Dullaghan 2021](#)). We are launching a [tournament on Metaculus](#) to incentivize more accurate and transparent predictions. This report lays out an initial set of forecasts and maps areas of uncertainty.

Two other things worth keeping in mind as you read this report:

- Consider the prospects of cultured meat relative to other things that could reduce suffering of farmed animals, especially if these are likely to reduce more animal suffering at a faster rate than cultured meat. This includes things like promoting other alternative proteins such as plant- or fungi-based meat, encouraging eating less factory farmed meat, animal welfare reform commitments from producers and sellers, legislation prohibiting the worst practices of factory farming, and genetic editing of animals to reduce their capacity for suffering.
- An implicit assumption in the theory of change here is that the reduction in animal suffering is because cultured meat production displaces conventional meat production. This may not be true, but we do not discuss it in this report.

Why the focus on production volume

Past work and forecasting questions have had different stakeholders and theories of change in mind. This project is interested in whether philanthropic dollars can be used to nudge cultured meat trajectories in a way that makes a significant positive impact on farmed animals. Others making claims about cultured meat are interested in additional impacts such as environmental

[Schwartz 2021](#)), or for cultured meat to be a premium specialty product or an ingredient in plant-based meats in the short to medium term ([Peppou 2021](#); [Swartz 2021a](#)). Few quantify what production volumes would satisfy these needs and this leads to some confusion about what counts as optimism or pessimism. Someone hoping to end factory farming in a few decades might see 100,000 metric tons as pessimistic, but someone hoping to make a profit selling cultured tuna might have a reason to be optimistic. For others, even millions of tons of cultured meat might be framed pessimistically if it does not bring with it a transformation of who has power in the food system.

Prior work on cultured meat forecasting has mostly focused on price targets, but we operationalized questions on either (price, volume) pairs or just pure volume. Claimed prices may be unreliable since venture capitalists can burn large sums of money on subsidies for small volumes ([selling at a loss](#)), so we believe that restaurant or storefront prices are a poor signal for ultimate cost-competitiveness. Other questions have asked about maximum production *capacity* of a single facility, but this only matters in worlds where the economics allow such capacity to be fulfilled, so we think it is less directly useful for the EA community than actual production volume.

We have intentionally chosen to define cultured meat products as being made mostly from cellular meat (>51% of the meat^[1] is from cultured animal cells) rather than plant-based meat with small cultured muscle or fat additions. It seems plausible the path of cultured meat production will be as an ingredient in otherwise plant-based meat ([Specht et al 2021](#); [Swartz 2021c](#); [Ben-Arye 2020](#)). Probabilities for production volumes of meat products that are <51% cultured meat might be higher. We did not forecast on them here but we include a question about $\geq 20\%$ cultured meat in the tournament. One theory of change is that investing in alternative proteins can reduce farmed animal suffering by speeding up the development of products that will displace conventional meat. A hypothesis in this theory is that “true” cultured meat products (not plant-cultured hybrids) will make up most of the products that people are willing to replace conventional animal meat with because they are closer to the original. If the EA community buys this, then forecasts of low production volumes may help raise awareness of the problems with this hypothesis and shift the EA community to consider other hypotheses more seriously.

Mapping uncertainty around speculative technological innovation is hard, especially when looking at timelines of 10, 15, and 30 years out. We initially began by reading the existing techno-economic analyses (TEAs), consultancy reports, and articles from the EA community and experts on cultured meat (discussed below).

Given our lack of domain knowledge in this area, we put a lot of weight on TEAs: CE Delft ([2021](#)), Humbird ([2020](#)), and Risner et al. ([2020](#)). While we do not have experience analyzing TEAs, there was a clear difference in rigor ([Zhang & Dullaghan 2021](#)) and we read the TEAs carefully enough to notice a [conceptual error](#) in CE Delft ([2021](#)) (which resulted in the authors [correcting the specific error](#) but not addressing more general concerns about their approach). We also had informal conversations with cultured meat scientists to get a sense of how plausible they found the claims made in the TEAs. Quoting from our review of the TEAs: “Humbird (2020) is very high quality and suggests cultured meat cost-competitiveness is hard and needs everything to go right. CE Delft (2021) outlines some of what will need to go right, but doesn’t provide much evidence that any of it is possible, has internal validity errors, and arguably has too much motivated reasoning. Risner et al. (2020) is decent, within the narrow limits it sets itself, but too many details are under specified for it to reflect the full costs and challenges of scaling up cultured meat” ([Zhang & Dullaghan 2021](#)).

We read through the public opinions of experts, think tanks, public intellectuals ([Broad 2021](#); [Datar 2021](#); [DeSantis 2021](#); [Dutkiewicz 2021](#); [Dyson 2021](#); [Fassler 2021](#); [Hayek 2021](#); [Mosa Meat 2021](#); [NewHarvest 2021](#); [Phillips 2021](#); [Peppou 2021](#); [Schweizer 2021](#); Shah [2021a](#), [2021b](#); Specht et al. [2021a](#), [2021b](#); Swartz [2021a](#), [2021b](#), [2021c](#), [2021d](#); [Open Philanthropy 2015](#)), and posts made on the Effective Altruism Forum about cultured meat ([Avacyn 2021](#); [Fish 2021a](#); Stijn [2021a](#), [2020a](#), [2020b](#); [Wen 2020](#)).

We developed forecasting questions on cultured meat reaching annual production volumes in metric tons (>100,000, >1M, >10M, >50M) before the end of a certain year (2031, 2036, 2051) with some iterations on price and species (<\$10/kg wholesale and cow-based meat). We also included variables we hypothesized would be signposts of progress (biotech venture capital and

retail and quick service restaurant sales, public protests, and public opinion surveys). Note these forecasts were all conditional on no [transformative AI](#) arriving before their resolution.

We (Neil and Linch) independently made forecasts on these 29 questions, discussed our results, and then made updates.

We also paid a panel of five highly ranked Metaculus forecasters and one cultured meat scientist to answer the questions we formulated. The forecasters were given from October 27, 2021 until November 15, 2021 to spend 15 hours making forecasts. [Here](#) is the recruitment, outreach, and payment used to assemble the forecasters, as well as the reference material provided to forecasters to optionally assist them in creating their forecasts. Forecasters were not able to see the predictions from others on the panel.

The forecasters' probabilities were transformed into odds and then we calculated the [geometric mean of the odds](#). Finally, we transformed these odds back into probabilities to arrive at the aggregate results per question.

Results

The spreadsheet of anonymized forecasts on all questions and the forecasters' reasoning is [here](#). The aggregated probabilities of cultured meat production targets from all the forecasters are given in the table below. For example, the aggregated forecast is a 15% probability that >100,000 metric tons of cellular meat (where >51% of the "meat" is produced directly from animal cells) will be sold at any price within a continuous 12-month span before the end of 2031.

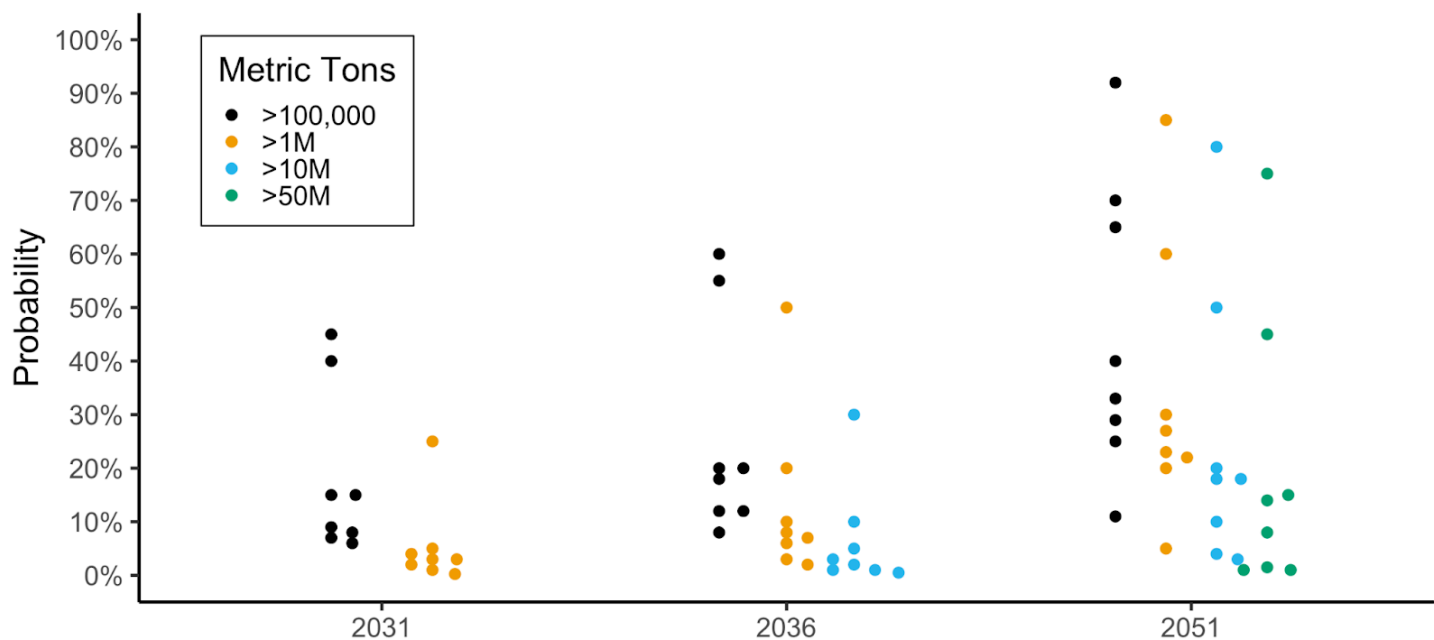
Aggregated probabilities of cultured meat production targets

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production in 2020 was 30,000 to 100,000 metric tons (the former according to Shapiro (2020) and this [paywalled page](#) from Meatingplace.com cited in Bollard (2020); the latter according to [data](#) obtained from FoodTrending.com). 13M metric tons of alternative protein (meat, seafood, milk, eggs, and dairy, excluding pulses, tofu, and tempeh) were consumed globally in 2020 (BCG 2021). ~545M metric tons of conventional [meat](#), including [seafood](#), is produced each year (according to OurWorldinData), mostly via the industrial farming of animals.

The average of all the differences between the highest and lowest probabilities for all the production volume questions is 48 percentage points, suggesting some variation. The figure below plots the probabilities per year for the pure volume questions (for the sake of simplicity, this graph does not show the probabilities for the questions about <\$10/kg meat or exclusively cow-based meat). While there was some variation in probabilities (especially 2051 probabilities), overall the probabilities for large volumes of cultured meat were below 50%.

Probabilities for metric tons of cultured meat that will be sold at any price within a continuous 12-month span before the end of the year

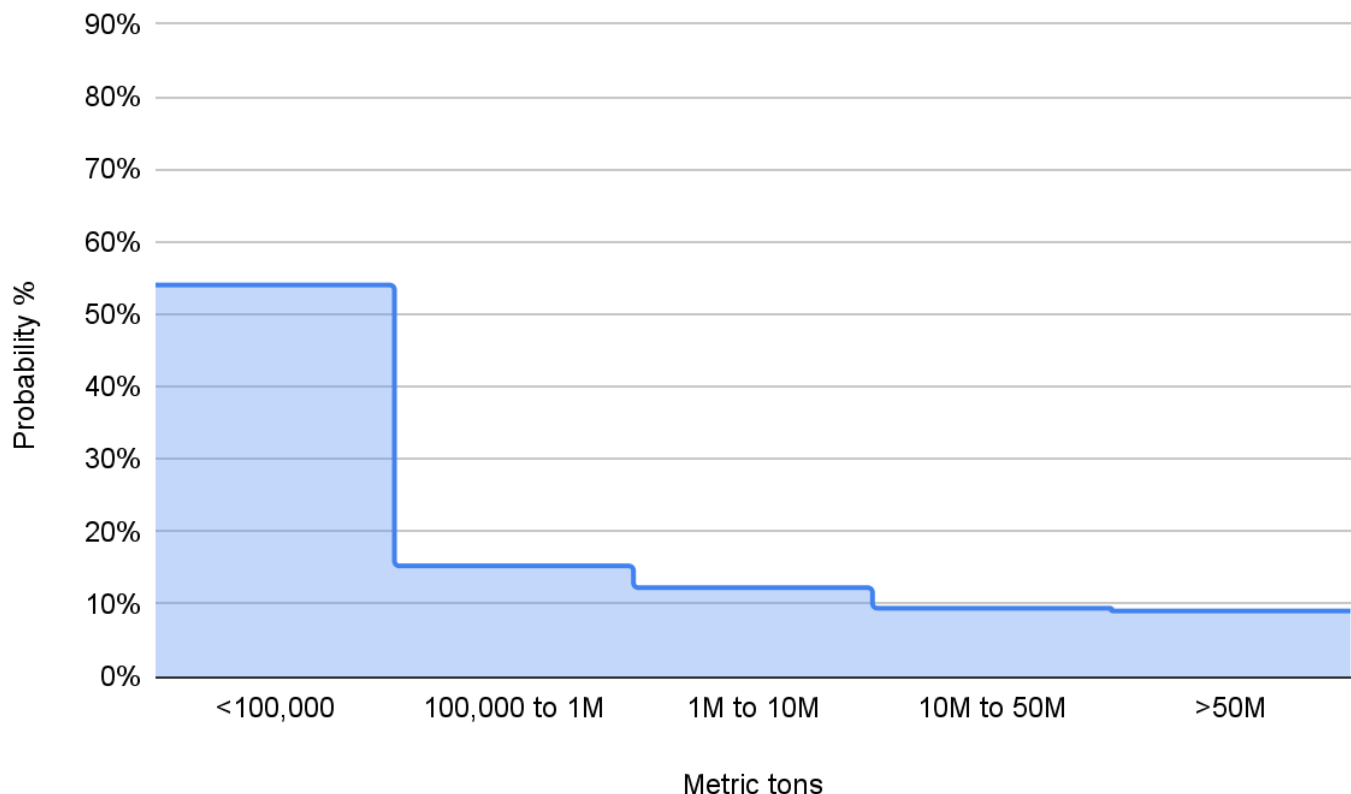


The results also show:

200,000 metric tons per year.

- The probability for >100,000 metric tons of cultured meat sold in a 12-month period before the end of 2031 was 15%, but the probability for >100,000 metric tons *sold for <\$10/kg wholesale on average* in a 12-month period before the end of 2031 was 6%.
- The probability for >100,000 metric tons of *cow-cell-based* cultured meat sold in a 12-month period before the end of 2031 was 8%, but the probability for >100,000 metric tons of *cow-cell-based sold for <\$10/kg wholesale on average* in a 12-month period before the end of 2031 was 4%.

Probability distribution for metric tons of cellular meat that will be sold at any price within a continuous 12-month span before the end of 2051



The full list of questions is available in the table at the end of this post.

The questions about signposts of progress also yielded some interesting results, including:

venture capitalists by ECFI were relatively low (20% China, 10% USA, 10% EU, 30% biotech venture capitalists).

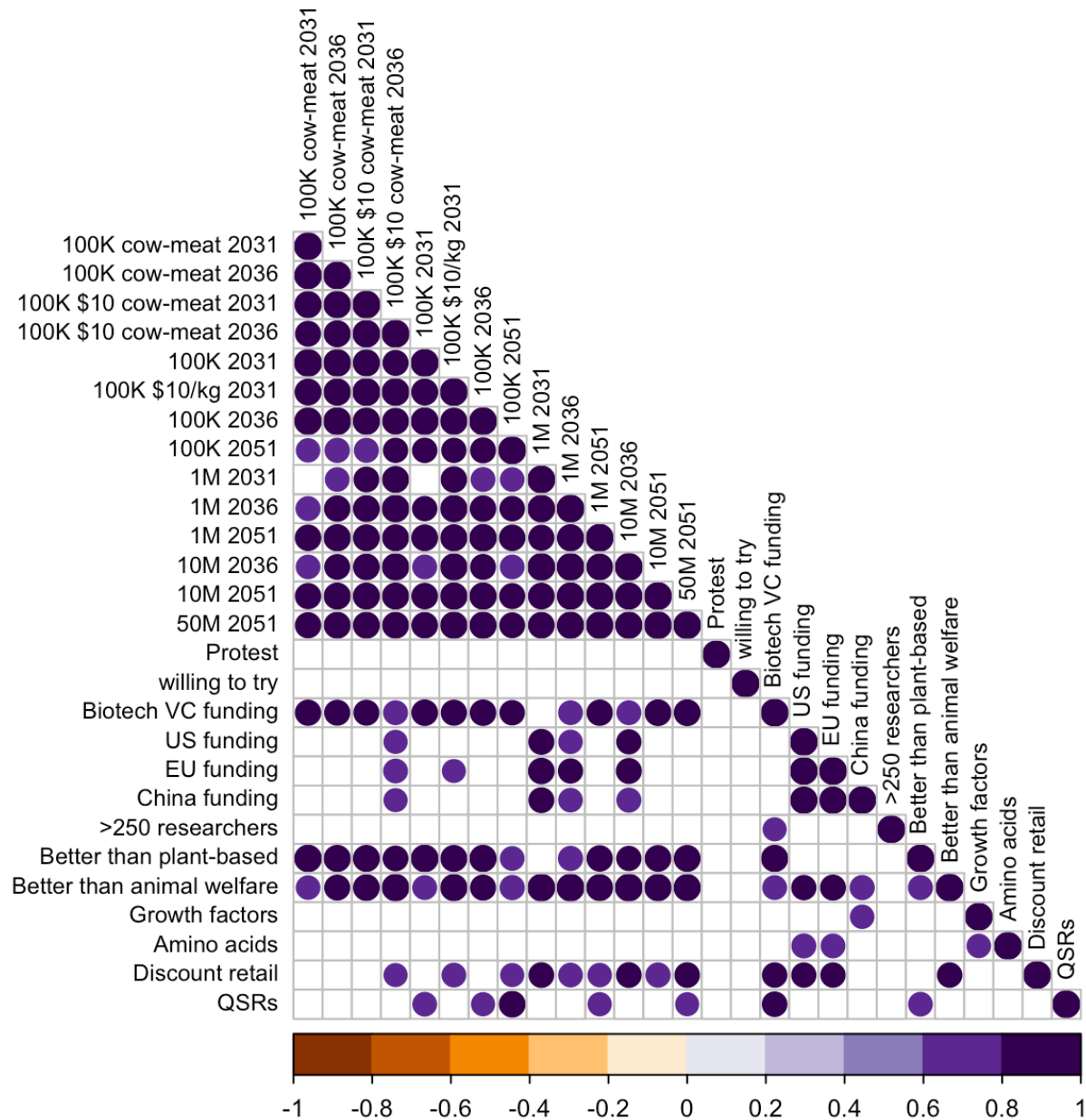
- The probability of there being more than 250 cultured meat researchers in 2036 was about even (52%). (At the time of forecasting, October 2021, the Good Food Institute (GFI) listed [37 cultured meat labs](#) and [60 researchers](#) open to working on cultured meat.)
- The probability of the largest discount store chains or quick service restaurants selling cultured meat by 2031 was low (8% and 19% respectively).
- Consumer approval did not appear as a major constraint as measured by the probability of large-scale anti-cultured meat protests (18%), or the majority of survey respondents saying they would be willing to try cultured meat (70%) (conditional upon >10,000 cumulative metric tons of cellular meat produced at any price by 2031).
- The probabilities that impartial effective animal advocacy evaluators in 2036 will view having spent \$10M this decade on cultured meat as better than on plant-based meat or increasing spending in a dollar-weighted average of Open Philanthropy's existing farmed animal welfare donations were 36% and 22% respectively.
- The probability for achieving cheap (<\$1/kg) growth factors by 2031 was 30% (growth factors are various hormones, cytokines, vitamins, and some other proteins that promote cell growth).

Analyzing the correlation of production volume predictions and predictions on the other inputs (media prices, public funding, biotech funding, and researchers) does not reveal much. However:

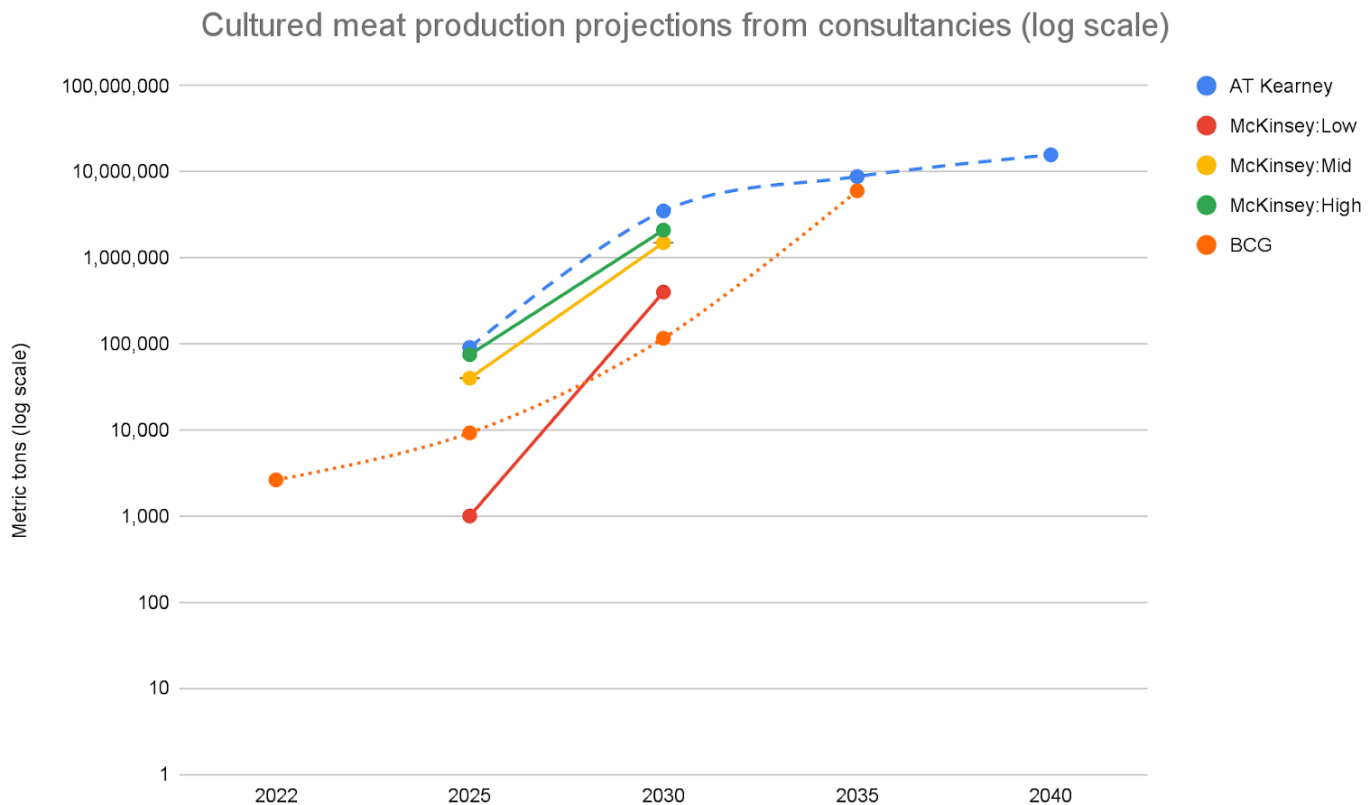
- The biotech venture capital prediction seemed correlated with many of the production volumes.

volumes (<1M before the end of 2031 and 2036, >1M before the end of 2036, >100K cow-meat for <\$10/kg before the end of 2031).

Other inputs were not significantly correlated with production volumes (numbers of researchers, consumer approval, cheap media). The figure below presents only the correlations reaching conventional levels of statistical significance ($p < 0.05$). No statistically significant negative correlations were found (data and script available [here](#)).



consultancies ([McKinsey 2021](#), [BCG 2021](#), [AT Kearney 2020](#)). They are also lower than what GFI expected if the cultured meat production life cycle followed the path of other industrial biotechnologies (suggesting millions of tons in the next 10 to 15 years (timestamped video [GFI 2021a](#) though GFI have not issued any official forecast or predictions on cultured meat production volumes)).



Source: [own calculations](#)

The calculations behind the growth rates proposed by these consultancies are not public and some of the claims they make are unsubstantiated, biased, or contradictory. For example:

- AT Kearney ([2020](#)) and McKinsey ([2021](#)) both assume amino acids can be scaled cheaply, which Humbird ([2020](#)) thought was unlikely.
- McKinsey ([2021](#)) leans on reference classes of genome sequencing and large-scale fermentation, but does not explain how those reference classes translate into their projections nor if they looked at other potential reference classes.

learning R&D could bring the cost down to ~\$11/kg but their market size estimates imply cultured meat products in 2030 will cost on average \$12-\$13/kg regardless of their low/medium/high growth scenarios.

- AT Kearney ([2020](#)) refers to expert opinion that prices will fall to \$40/kg by 2032, but otherwise writes very little to justify the estimates it proposes.
- BCG ([2021](#)) leans on a survey of experts who point to the possibilities arising from metabolic efficiency, cheap media, and unstated “key non-muscle-meat ingredients,” but do not discuss these in detail.

Given the few significant correlations between our hypothesized input factors (media, researchers, consumer approval, and media costs), we discuss the forecasters’ reasoning more qualitatively below to tease out the cruxes behind their forecasts. We encourage interested readers to examine some of the detailed and well-considered reasoning given by forecasters.

The cruxes that appear to shape the reasoning for our forecasting panel are their probabilities that:

- Cultured meat is a case where technological innovation can outperform the efficiency of the biological system it is trying to replicate.
- Cultured meat still faces difficult fundamental challenges.
- Cultured meat is analogous to high-growth reference classes, especially ones that were supported by government funding (such as photovoltaics and genome sequencing).
- Humbird ([2020](#)) is wrong about some fundamentals, especially media costs.

Efficiency of innovation

A stylized version of the most optimistic argument that cultured meat technology will scale is: “There are challenges, but with time and money, technological innovation can replicate or outperform any biological system” —

genome editing. The optimistic argument is that it is only a matter of growing the one part of the animal you want without all the extra unnecessary bits (brains, ears, etc.) and no rules of nature need to be violated, so it should not be difficult to be more efficient ([Stijn 2021b](#); [Hayek 2021](#)).

The pessimistic argument is that the task of growing cultured meat is to replicate each of the parts of cell growth (immune system, cell differentiation, nutrition supply) and make them compatible in a single process, but this is unlikely to be the approach that gets you meat more efficiently (and cheaply) than factory farming ([Fish 2021a](#)). This pessimistic argument says technologies that replaced biological systems (solar panels vs. plants, cars vs. horses, planes vs. birds, recombinant vs. porcine insulin) were new (and better) ways of accomplishing a goal that were completely free from the limitations of the biological systems they've replaced ([Fish 2021b](#); [Zhang 2021](#)).

The range of these opposing viewpoints are exemplified by our forecasters:

- Forecaster 2 held "A general view that anything which is possible in nature should be replicable on some level by humans with sufficient effort."
- Forecaster 3 thought some of the innovation challenges were likely failure modes: "Cells with optimized metabolism or improved characteristics to meet other constraints will need to be developed (40% failure). The factory and bioreactors will need to be kept sterile to avoid contamination by microbes and virus, and it's an open question on whether this can be done cost-effectively, and there are downstream challenges caused by sterilization ("Heat-stable media" to withstand HTST sterilization) (25% failure)."
- Forecaster 4 wrote, "Cultured meat is mimicking what's happening in an existing pretty well-optimized bioreactor (animal)."

Difficulty of challenges

The GFI state of the industry report ([2021b](#)) claims "fundamental technological breakthroughs are not necessary to eventually achieve economically viable, scaled production of cultivated meat," but they have also stated, "there are some

thermodynamics, and other constraints ([Specht et al. 2021a](#)). They argue that given the nascency of the field, we are not close to exhausting opportunities to innovate around problems ([Specht et al. 2021b](#)). However, they do not identify these “true constraints” or suggest why they would not also constrain a more mature field.

- Forecaster 2 wrote, “Whilst there seem to be high technical challenges ahead, the fact that we already can successfully grow [cultured meat] means it seems unlikely to me we will be able to find a blocker (beyond economics, which isn’t super compelling yet).”
- Forecaster 1 thought >100,000 metric tons per year before the end of 2051 “seems quite likely (92%) unless the development of the technology encounters nuclear fusion levels of difficulty.”
- Forecaster 6 wrote, “it’s moderately likely that fundamental biological limits come into play that make cultured meat look harder than the manhattan project, which was made difficult by practical, but not theoretical/fundamental, issues,” while also noting that >100,000 metric tons per year before the end of 2031 sold for <\$10/kg wholesale “would require transformative breakthroughs in more than one area, which is very unlikely, though not impossible.”

Reference classes

Those bullish on cultured meat often point to price drops in solar panels and batteries as evidence that cultured meat can scale. Those who are bearish often point to the limited production of genetically modified (GM) crops and biofuels.

However, these arguments do not tell us *why* those cases succeeded or failed and why cultured meat is more like those cases than ones where the growth trajectory was different. To overcome this limitation:

- We should draw on relevant causal models that could explain whether or not an innovation overcomes the various obstacles in cultured meat’s path. (Some possibly useful models are described by Alexander ([2017](#)), Bolsch & Fenn ([2018](#)), Colbin ([2021](#)), Crawford ([2021a](#), [2021b](#)), Cristensen et al.

Smith ([2017](#)), Hammer ([2017](#)), Hetherington & Ross ([2015](#)), Hetherington (2017), and Ritchie ([n.d.](#))).

- Pessimists and optimists should more clearly explain what they think the causal mechanism was in their preferred reference class, and why it is the best analogy to mechanisms we see with cultured meat. Is there strong evidence that cultured meat has the same causal mechanism that led solar to grow rapidly? Or is the causal mechanism that limited GM crops and biofuels more supported by the available evidence? We need to exchange models and concrete predictions.

Across the series of questions, the forecasters in our panel made references to nuclear fusion, photovoltaics, insect-based foods, GM foods, plant-based meat, and nanotechnology.

- Forecaster 5 thought that >10M metric tons before the end of 2036 “seems very unlikely, solar didn’t grow at the rate required for this (assuming we get to [100,000 metric tons] around 2028 say) despite getting exponentially cheaper and a very favorable political environment including lots of subsidies.”
- Forecaster 4 wrote, “one might say, what about genome sequencing — I think genome sequencing is much more expensive than relevant cellular processes. It doesn’t do anything sufficiently novel to unlock great powers (as opposed to a nuclear chain reaction for explosives or solar energy for energy).”
- Forecaster 3 wrote, “I think a comparison to various research fields would be useful here, both rapidly growing fields as well as field where the hype had died down. Nanotechnology as a whole has more than 1000s of researchers; a web search turned up a directory of 1800 researchers. Perhaps the comparison would be to some subfield of nanotechnology, for cultured meat. If compared to some subfield of machine learning, it should be easy to clear 250 researchers.”

The table below shows the compound annual growth rate (CAGR) needed to reach production targets, assuming cultured meat production in 2021 was 1 or 10

hundreds of kilograms).

CAGR needed for cultured meat projections

If cultured meat annual production in 2021 is, in metric tons:	1	10
Annual metric tons target	CAGR needed	
	2021-2031	
100,000	216%	151%
1M	298%	216%
	2021-2036	
100,000	115%	85%
1M	151%	115%
10M	193%	151%
	2021-2051	
100,000	47%	36%
1M	58%	47%
10M	71%	58%
50M	81%	67%

- Forecaster 3 wrote, "If [100,000] metric tons could be sold in 2031 for \$40/kg, then at 60% growth every year 1M metric tons at \$20/kg can be reached in 2036. The former probability for 2031 is quite low, but the conditional is not too bad, maybe 50% for that kind of growth."
- Forecaster 5 wrote that achieving >1M metric tons in a 12-month period before the end of 2051 "is just "hit [100,000 metric tons] by 2045 and grow at 15% annualised minus reasons the product doesn't become unpopular in the interim which seem unlikely to me" (27%).

were looking at something exponential, but from a very low base. I think the McKinsey / ATK analysis is pretty optimistic and slightly weak. I looked at the various estimates from the companies linked in the intro document, and think they are plausible, but downweighted them a little to assume a slight optimism bias, but I think generally they aren't crazy."

- Forecaster 6 thought that building out many facilities as large as Good Meat's planned 4,500 metric ton per year plant in Doha or a smaller number of larger facilities was unlikely in the timeframes needed.

One attempt to map out the CAGR in seven reference classes that are often touted as analogous to cultured meat (production of solar energy, electric cars, GM crops, biofuels, margarine, nuclear fission, and lithium-ion batteries) will be made in a separate write-up on Metaculus (Dullaghan 2022, *forthcoming*). We encourage others to clearly quantify the growth rates of their proposed reference classes.

Reliance on Humbird (2020)

As most of the forecasters were not domain experts, they were required to make a judgment call on how much weight to put in the more pessimistic estimates from the Humbird ([2020](#)) TEA versus other more optimistic signals.

- Forecaster 1 (the most optimistic of the panel) made no reference to Humbird's estimates, and wrote, "agree with Specht (2019) that cultivated meat is likely to be viable in the long run, so the question is mainly how fast the first successful start-up will ramp up production."
- Forecaster 2 claimed they "found the techno-economic analyses compelling, and despite alternatives being discussed, it seems at least 50% likely they are "broadly correct". I'm roughly 60/40 in the case where they aren't broadly correct / there's some weak evidence of things not being considered, and the \$ in the space are compelling." This led them to revise down their otherwise higher probabilities, which were based on priors that humans can replicate anything in nature with sufficient effort and a sense that "people are investing real \$ into this space in order to achieve it."

challenges outlined by Humbird (2020). When estimating >100,000 metric tons of cow-meat by 2031 they wrote, “there should be enough capacity to produce that much: maintaining medical grade facilities as outlined in Humbird feels quite crazy. So this prediction bets on us making a breakthrough with bioreactors.” When estimating >100,00 metric tons of cow-meat by 2031 for <\$10/kg they wrote “Humbird’s report makes me think that this might not be possible (if we are talking about plain meat without additives). I buy into his pessimism. And flushing VC money to subsidize production might not happen if there are fundamental reasons against feasibility.”

- Forecaster 6 thought “there is some chance that an innovative technology (e.g. cell lines that produce their own growth factors, engineered growth factor signaling pathways that respond to cheap small molecules) make this possible” but that improvements above Humbird would require scaling that was unlikely to happen in the given timeframe.

While some forecasters referenced the obstacle of sterile bioreactor systems, few made reference to Humbird (2020) at the same time. This was somewhat surprising since our read of the TEAs was that the [Food versus pharmaceutical grade](#) assumptions could be a key source of uncertainty and error (some forecasters did note that the costs of factories in Humbird (2020) CE Delft (2021) were not that different so the pharmaceutical-v-food grade system issue did not seem to be a big crux). Instead, the reliance on Humbird (2020) was especially true regarding how forecasters thought about costs of growth factors and amino acids.

- The pessimistic argument is that pharmaceutical-grade amino acids are the only input meeting the required purity and efficiency needs, but are limited in supply and costly.
- The optimistic argument is that the industry is not aiming to produce and purchase pharmaceutical-grade amino acids individually, but instead to mostly use widely available cheaper food- or feed-grade alternative sources, which Humbird did not consider realistic ([Mosa Meat 2021](#); [Swartz 2021c](#)).

transform and include in the reference material for the amino acid question. As these are unrelated and more expensive, it is possible this biased forecasters' estimates. None of the forecasters seemed to notice this error in their reasoning.

- Forecaster 3 wrote, "Cost of amino acids may possibly be much lower than as reported by Humbird, as existing food-grade amino acids are available at \$30/kg" but used Humbird's \$19.23/kg estimate as a lower bound when giving a 50% probability "that food-grade amino acids will be sufficient for cellular meat production."
- Forecaster 4 viewed cheap amino acids as unlikely (10%) due to the Humbird ([2020](#)) TEA.
- Forecaster 5 thought our \$1/kg growth factor cost "is much cheaper than Humbird suggests, but I don't feel like the case for not achieving it was made convincingly and find it hard to have much confidence one way or another" and gave a 35% probability to such cheap growth factors by 2031. Our \$20/kg amino acids cost was just above the Humbird (2020) estimate and Forecaster 5 wrote, "I think there's some chance e.g. the Nutreco feed grade solution or some similar thing is possible" and gave a 40% probability to cheap amino acids by 2031.
- Forecaster 6 explicitly stated, "my crux here is the potential development of an alternative amino acid source (e.g. low cost plant hydrolysates) that does not require sourcing them individually," and gave a 14% probability to cheap amino acids by 2031. On growth factors, they wrote "this would require significant improvements over Humbird's predictions, which are already predicated on a scale of production that I don't think will be reached in this timeframe. However, there is some chance that an innovative technology (e.g. cell lines that produce their own growth factors, engineered growth factor signaling pathways that respond to cheap small molecules) make this possible," and gave a 6% probability to cheap growth factors by 2031.

Innovations needed

tens per year before the end of 2021, what can we do? Based on our review of the TEAs and the cruxes identified by our forecasters, we should focus on:

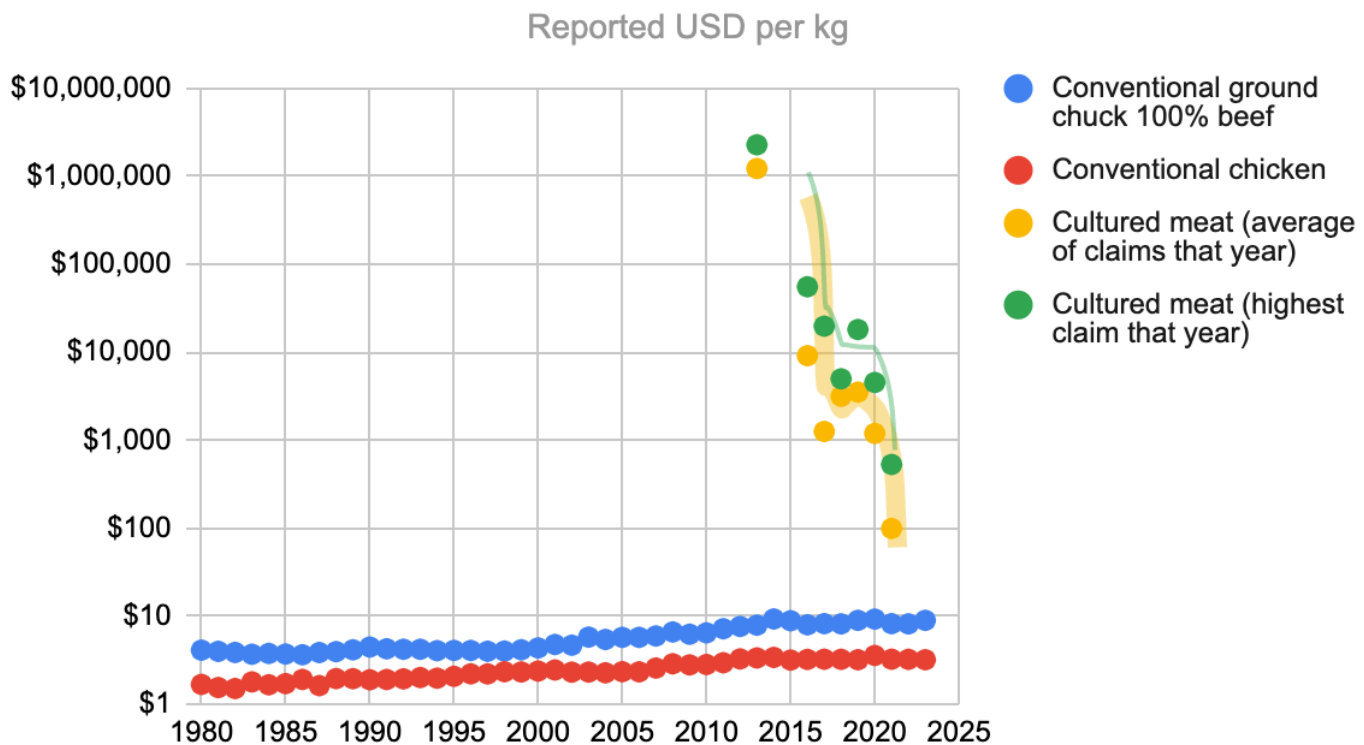
- **Engineering new types of bioreactors** that are: large enough (tens of thousands if not 100,000 liters — up from ~1,000–10,000 today) and cheap enough (reduced from ~\$778,000 for 20,000 liters to ~\$60,000) without the costly sanitary requirements of pharmaceutical cleanrooms, and able to efficiently cycle out waste. (One could alternately have many smaller bioreactors but they would also need to be cheap and easily sanitized)
- **Lowering the costs of amino acids** by either: replacing pharmaceutical-grade with food-grade standards (or even feed-grade), or circumventing the need for pure amino acids by using plant hydrosolates to bring down the cost from hundreds of dollars per liter to <\$1/L — and then scaling up this ingredient industry.
- **Attracting billions in annual public funding** to pay for R&D and infrastructure grants/loans to survive any potential private investment winter or intellectual property barriers.
- **Developing innovations in cell engineering** to improve the media consumption rate (such that cells need about 10 times less media), and allow cell density limits to be above 25% (i.e., more meat per batch).
- **Lowering the cost of growth factors** from millions of dollars per gram to <\$1/g.
- **Increasing the CAGR of researchers** in the field to more than 10% for a decade.

What would change our probabilities

Below we list what would make us update, and in the sub-bullets offer some information possibly meeting that update criteria. Lynch's list was made before

Linch (2021 October 9)

- **Social proof.** For example, if the top biotech or most prestigious traditional venture capitalists made major investments, or if top ~10 Metaculus forecasters or other top forecasters believe something very different after spending considerable time looking into this.
 - Two of our Metaculus forecasters were more optimistic than the others, despite all being highly ranked Metaculus users.
 - The aggregate probability from our forecasting panel was a 35% probability that at least two companies in our list of top biotech venture capitalists (Arch, Flagship, Atlas, Third Rock) would each lead a funding round of total size >\$100M in a cultured meat company, or credible evidence of >\$3B total in funding rounds for cultured meat companies (with participation from at least two of the predefined top biotech venture capitalists) before the end of 2031.
- **Prices below \$100/kg.** Humbird's ([2020](#)) analysis claims costs of >\$200/kg using wild-type animal cells, so credible evidence of <\$100/kg in the near future (say, the next three years) would be a moderate refutation of this model, even though his endline numbers were lower.
 - There are already such claims from Shiok Meats ([\\$50](#)) and Future Meat ([\\$68](#)). It has been [claimed](#) Avant Meats can produce at \$39/kg. Future Meat also claims to be able to produce a product for \$17/kg ([Mitch 2021](#)) (but it might be closer to \$30/kg when including the cost of buildings, construction, equipment, installation, labor, and other factors ([Charles 2021](#))).
 - [Reported prices](#) appear to be falling quite rapidly (as shown in the chart below).



- **Amino acid and other input prices coming down.** Cost estimates from the TEAs we've read suggest growth factors cost hundreds to billions of dollars per gram, and amino acids cost thousands to hundreds of thousands of dollars per gram.
 - The animal feed company Nutreco claims their feed-grade amino acids will work just as well as pharmaceutical-grade (L-lysine costing €3/kg for feed-grade versus €227/kg for pharmaceutical-grade) ([Good Food Conference 2021](#)).
 - Upside Foods claims to be using plants milled into powders (instead of lab-grade amino acids) ([Upside Foods 2021](#)) and food-system-scalable, animal-free media (amino acids, growth factors, proteins, etc.) ([Watson 2021](#)).
 - Nordic Virtual Pastures claims ([2022](#)) "We purify grass remains from harvested fields to create nutrients from which bovine cells grow in bioreactors. This greatly reduces the price of cultured beef, yielding a more sustainable alternative to conventional beef."
 - Super Meat [claims](#) to use a serum-free medium formulation that costs \$2/L.

expected the average cost of medium to fall to over \$5/L by December 2021. 63% thought the lowest production cost of media they can achieve in the next 5 years (by 2026) would be below \$5 (37% expected it to be higher) ([GFI 2020](#)).

- Laurus Bio is targeting \$1/g for albumin and transferrin and \$100/g for growth factors ([Good Food Conference 2021](#)).
- Mosa Meat claim their animal-free media now performs on par or better than fetal bovine serum-based media at increasingly lower cost (since 2019) ([2021](#)). However, their serum-free media did not achieve 3D cell differentiation comparable to that of serum-starvation control conditions ([Southey 2022](#)). “Thus, the serum-free media will need further optimisation for a 3D culture system, as well as market-ready biomaterials that could be scaled to industrial food-grade production levels” ([Messmer 2022](#)).
- Venkatesan et al. ([2022](#)) showed the use of newly discovered growth factors in Essential 8 media reduced its cost 11-fold and reduced the cost component of growth factors from 95% to 4%.

- **Direct or indirect evidence that cheap hygiene at scale is much easier than Linch thinks (heavily influenced by Humbird estimates).**

- Mosa Meat ([2021](#)) said they'll use pharmaceutical-grade cleanrooms for some parts of the facility, but the majority will not require it. Instead, bioreactors in a food-grade space will provide sufficient closed environments for sterility controls, and this can offset major costs.
- Matthew Stork [believes](#) single-use bioreactors (effectively big plastic bags) can be produced and used more cheaply than fixed stainless steel reactors. Humbird [believes](#) these would still be costly due to set-up and tear-down costs and absolute limits on liter size of such single-use bioreactors.

- **Conceptual/analytic framework.** We haven't seen anyone offer a knock-out response to the argument that trying to replicate biological systems

approaches.

- **If reasonable people already did a comprehensive, first-principles-based search on ways to reduce animal suffering and landed on cultured meat and plant-based meat as the best methods they can find.**
 - As far as we know, GFI and New Harvest didn't do this.
 - Other alternative protein candidates seem plausible ([Zhang 2022](#)), like less well-known Chinese tofus ([Stiffman 2022](#)).

Neil (2022 March 1)

My personal take is that we could have >100,000 metric tons of cultured meat per year sold at any price in 15 years if an ultra-wealthy business entrepreneur (for example, Elon Musk) made it their goal and poured in billions of dollars per year while making a huge financial loss. The overriding obstacle I see is scaling to millions or tens of millions of metric tons requires an enormous and expensive infrastructure and ingredient supply build-out — and that this is not necessary for cultured meat companies to make a profit. I see similar challenges for plant-based meat but without the concerns around pharmaceutical costs. Having said that, I am not an expert and EA has a lot of money. I'd be willing to put up to \$10 million into lobbying/prizes for public R&D on fundamental breakthroughs in material science, sterility, and hydrosolates which will survive any private investment winter.

I would update these claims based on the following:

- If David Humbird [visited the Upside Foods facility](#) and said afterward that his TEA was too conservative about the possibilities available from innovation.
- Public demonstrations of large batch production (>1,000 metric tons) in the next two years. They don't even have to be tasty, just free from >15% batch contamination.

showed cost reductions below Humbird's ([Lee 2022](#)).

- If the results of a Breakthrough Institute-backed [study](#) of cultured meat infrastructure needs in Thailand or GFI's upcoming infrastructure analysis of cultured meat ([Specht 2022](#); [Swartz 2022](#)) included credible evidence that Humbird's ([2020](#)) estimates of infrastructure needs were too high.
- A government-sponsored program worth >\$500M per year to produce feed-grade macronutrients specifically for cultured meat, or loans/grants for novel bioreactor build-outs.
 - China included cultured meat in their official [Five-Year Agricultural Plan](#) ([Vegconomist 2022](#)), which refers to cell-cultured meat as part of food manufacturing of the future.
 - In October 2021, the United States Department of Agriculture awarded Tufts University [\\$10M](#) to establish the National Institute for Cellular Agriculture, and in September 2020 awarded [\\$3.5M](#) to researchers at UC Davis for a five-year program to find cell lines and cheap plant-based media.
 - Singapore's A*STAR Research [announced](#) a dedicated "CentRe of Innovation for Sustainable banking and Production of cultivated Meats (CRISP Meats)" focused on public-private partnerships.
 - South Korea has also recently established the Korean Society for Cellular Agriculture, which includes research on cultured meat ([Lee 2022](#)).
- Credible evidence that the maximum achieved percent of volume that cells make up in a bioreactor of at least 20,000 liters was above 25%, suggesting Humbird was wrong about some fundamentals.
 - Future Meat claims to be able to achieve 36% cell density in an 800-liter facility ([Poinski 2021](#)). They also claim to have achieved cell densities that are 10 times higher than competitors (i.e. 100B cells per liter) and developed a process that allows cells to grow as single-cell

surface area of their bioreactor ([Falcon 2021](#)).

- Credible evidence of large batches of cultured meat (>1,000 metric tons) produced in bioreactor systems without pharmaceutical cleanrooms.
 - Mosa Meats ([2021](#)) wrote that the bioreactors can be set up in a clean non-classified area, as these are completely closed processes. The seed stage of the bioreactors will be in a Class 6 ISO cleanroom, while the meat harvesting process and the transfer between proliferation and differentiation stages would likely be in a Class 8 ISO cleanroom.

Aggregated probabilities for all questions

Question	Aggregated probability
>100 metric kilotons of Cow-cell based cellular meat will be sold at any price within a continuous 12-month span before the end of 2031.	8%
>100 metric kilotons of Cow-cell based cellular meat will be sold at any price within a continuous 12-month span before the end of 2036.	15%
>100 metric kilotons of Cow-cell based cellular meat will be sold for <\$10/kg wholesale on average within a continuous 12-month span before the end of 2031.	4%
>100 metric kilotons of Cow-cell based cellular meat will be sold for <\$10/kg wholesale on average within a continuous 12-month span before the end of 2036.	7%
>100 metric kilotons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2031	15%
>100 metric kilotons of cellular meat will be sold for <\$10/kg wholesale on average within a continuous 12-month span before the end of 2031.	6%

price within a continuous 12-month span before the end of 2036	
>100 metric kilotons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2051.	46%
>1M metric tons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2031	3%
>1M metric tons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2036	9%
>1M metric tons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2051.	31%
>10M metric tons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2036	3%
>10M metric tons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2051.	18%
>50M metric tons of cellular meat will be sold at any price within a continuous 12-month span before the end of 2051.	9%
Conditional upon >10 cumulative metric kilotons (10,000 metric tons) of cellular meat produced at any price before the end of 2031, will there be large-scale protesting (at least 100,000 people within a 14 day period, including one protest of at least 10,000 people, within national boundaries of the USA, China, India, UK or an EU country) explicitly against cellular meat products, according to credible media reports (Agence France-Presse (AFP), Associated Press (AP), BBC News, Reuters and EFE)?	18%
Conditional upon >10 cumulative metric kilotons (10,000 metric tons) of cellular meat produced at any price, will >50% of a representative sample of US,	69%

to a survey group/research institute deemed credible by Open Philanthropy?	
Will at least two companies in our list of top biotech Venture Capitalists (Arch, Flagship, Atlas, Third Rock) each lead a funding round of total size >\$100M USD in a cellular meat company, OR credible evidence of >\$3B USD total in funding rounds for cellular meat companies with participation from at least two of the predefined top biotech VCs before the end of 2031?	30%
Will the US federal government have allocated at least \$1B across a 12-month period to fund cellular meat research before the end of 2031?	15%
Will the EU have allocated at least \$1B across a 12-month period to fund cellular meat research before the end of 2031?	13%
Will official Chinese national government sources claim to have allocated at least \$1B across a 12-month period to fund cellular meat research before the end of 2031?	25%
Will there be more than 250 PIs and PhD students full time equivalents working explicitly on cellular meat in 2036 (working in a lab, degree, or project dedicated to cellular meat), according to sources Open Philanthropy deems are credible?	52%
"Impartial Effective Animal Advocacy evaluators" (a majority opinion among Kieran Greig, Karolina Sarek, Alexandria Beck, Mikaela Saccoccio) in 2036 will consider that a marginal \$10M donated between 2021 to 2026 would have been better spent on cellular meat rather than plant-based meat	36%
"Impartial Effective Animal Advocacy evaluators" (a majority opinion among Kieran Greig, Karolina Sarek, Alexandria Beck, Mikaela Saccoccio) in 2036 will consider that a marginal \$10M donated between 2021 to 2026 would have been better spent on cellular meat rather than increasing spending in a dollar-weighted	22%

Will reputable sources estimate that growth factors (inclusive of but not limited to FGF2, TGF- β) contribute less than \$1/kg wet mass on average to the production of cellular meat in 2031?	30%
Will reputable sources estimate that amino acids other than growth factors (inclusive of but not limited to the form of individual amino acids, vegetable hydrolysate, and recombinant proteins) contribute less than \$20/kg wet mass on average to the production of cellular meat in 2031?	Due to the inclusion of “recombinant proteins” in the question and references to transferrin and insulin in the reference material, cost estimates here may have been incorrectly biased upwards.
Will one of the top ten largest discount store chains by revenue (akin to today's Target, Aldi, Lidl, Trader Joe's, Save-A-Lot Grocery Outlet, Costco, Walmart) sell cellular meat own-brand goods at any price before the end of 2031, according to credible industry reports?	8%
Will a well-known US Quick Service Restaurant (Subway, McDonald's, KFC, Pizza Hut, Burger King, Domino's, Hunt Brothers Pizza, Taco Bell, Wendy's, Hardee's, Chick-fil-A, Chipotle, Little Caesars, Dunkin, Starbucks, Baskin Robbins) offer a cellular meat product at any price before the end of 2031?	19%
In 2051, would cellular meat look harder in 2021 than the Manhattan project looked in 1942?	Due to an error discussed in the read me doc , we don't have full results for these questions
In 2051, would we believe that a Manhattan project for cultured meat in 2021 would have been successful?	
What will total global conventional meat and seafood production be in 2051 in millions of metric tons (80% confidence interval i.e. 80% probability that the true value is between this range).	Minimum was 30M and maximum was 1.5B

Credits



RETHINK PRIORITIES

This report is a project of [Rethink Priorities](#).

It was written by Neil Dullaghan. Thanks to Linch Zhang, Jacob Peacock, Jason Schukraft, and Peter Wildeford for their extremely helpful feedback and contributions. Any mistakes are my own. Thanks to the forecasters who agreed to be on our panel and to the cultured meat scientists who spoke with us. Thanks to Stefan Schubert who provided comments that improved the summary.

If you like our work, please consider [subscribing to our newsletter](#). You can see more of our work [here](#).

1. Note that the instructions given to the forecasting panel did not specify the 51% was “by weight”, which is what we intended. Nothing in the reasoning provided by forecasters reveals whether they were thinking about the 51% in terms of weight or volume.

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